

Biomolecular Condensates for Regulation of RNA Equilibria and Signal Transduction

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Abstract: The phase transition of proteins and RNAs, forming biomolecular condensates, has emerged as a key mechanism for organizing biochemical reaction networks. While multiphasic condensates are thought to modulate RNA function in vivo, the underlying physicochemical mechanisms remain difficult to probe in complex cellular environments. Here, we utilize oligopeptide-based coacervates as synthetic models to investigate how multiphase architectures alter RNA chemistry.^{1,2} By applying a four-state thermodynamic framework, we demonstrate that these compartmentalized systems exhibit synergistic RNA duplex destabilization. This phenomenon arises from the energetic coupling between phase-specific partitioning and hybridization equilibria, where the multilayered structure provides a chemical environment distinct from its individual components. Furthermore, I will discuss how the phase transition enables information processing within cellular signaling pathways by operating as non-equilibrium systems governed by kinetic balance and stochastic fluctuations^{3,4}. Using a novel single-molecule approach, we reveal stochastic nucleation behaviors that are obscured in ensemble measurements. These findings suggest that cells may exploit fluctuations to filter noisy signals and enable robust biological control. Collectively, these studies establish a thermodynamic foundation for molecular regulation within condensates.

References

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